Nanyang Technological University
AY2024-25, Semester 1
MH2100 (Calculus III)

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Basic Information About the Course

Course coordinator: Leonard HUANG (Dr).
Email address: leonard.huang@ntu.edu.sg.
Office: SPMS-05-08.
Office hours: Thursdays, 14:00 − 15:00.
Tutors:

- Leonard HUANG: DSAI3.
- Luo YI (Anthony): DSAI1, MAS1, MAS6, MAS8.
- NG Kai Xiang (John): DSAI2, MAS3, MAS5, MAS7.
- Alaric POW Ian-Jun: MAS2, MAS4.
- YANG Yanting: MAS9, MAS10.

Lecture Schedule and Important Dates

Lecture schedule (Weeks 1 − 13)

Wednesdays:	13:30 – 14:20, in LT27.
Thursdays:	16:30 − 18:20, in LT19A.

Important dates

Midterm Exam 1:	September 19, 2024 (Thursday), 16:30 − 18:20.
Midterm Exam 2:	October 24, 2024 (Thursday), 16:30 − 18:20.
Final Exam:	November 26, 2024 (Tuesday), 13:00 − 15:00.

Assessment Metric and Exam Policies

Assessment metric

Midterm Exam 1 (CA):	25%.
Midterm Exam 2 (CA):	25%.
Final Exam (FA):	50%.

Help-sheet policy

For all exams, every student is allowed to consult just one double-sided A4-sized help-sheet with only handwritten notes.

Make-up policy

NO make-up opportunity will be made available for Midterm Exams 1 and 2.

Missing a midterm exam without a valid medical excuse will result in a 0 score for it.

What Is Calculus III?

Calculus III is the generalization of calculus for real-valued functions of one real variable to real/vector-valued functions of several real variables, which are called "multivariate functions".

You should be familiar with single-variable calculus by now.

In Calculus III, we will learn the following higher-dimensional techniques:

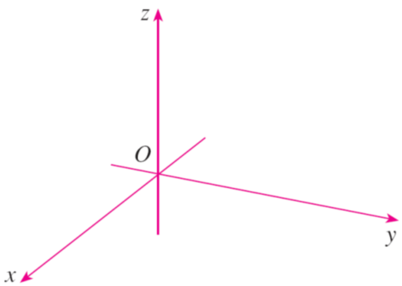
- Finding the volume of a geometrically defined solid in $\mathbb{R}^3$.
- Optimizing a real-valued function of several real variables. lagrange multipliers
- Integrating a vector field along a 1-dimensional path inside $\mathbb{R}^n$.
- Integrating a vector field over a 2-dimensional surface inside $\mathbb{R}^3$.

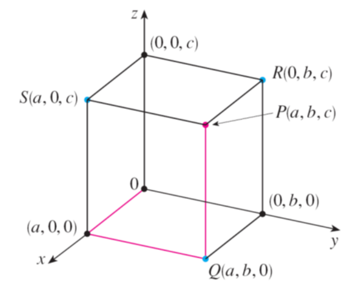
What Is a Point in 3-Dimensional Space?

A point in 3-dimensional space is just an element of 3-dimensional space.

Any point in 3-dimensional space can be represented as an ordered triple of real numbers, as we will see on the next slide.

Cartesian Coordinates in Dimension 3





The origin O: Any point <u>of your choice</u> in 3-dimensional space.

The Cartesian coordinate axes: Any three lines <u>of your choice</u> that are perpendicular to each other and intersect in the origin O. These lines are arbitrarily labeled as the *x*-, *y*-, and *z*-axes, but when they are drawn, they are oriented using the Right-Hand Rule.

Cartesian coordinates: Any point P in 3-dimensional space can be represented as an ordered triple (a, b, c) of real numbers, called "Cartesian coordinates", as follows:

- Uniformly assign real-number rulers to all three Cartesian coordinate axes.
- Then let a, b, and c be the real-number measurements of the perpendicular projections of P onto the *x*-, *y*-, and *z*-axes, respectively.

What Is a Vector in 3-Dimensional Space?

We can define an equivalence relation on the set of all directed line segments in 3-dimensional space by the following rule: Two directed line segments in 3-dimensional space are equivalent if and only if they have the same magnitude and direction. every vector can be "coordinatized"

A vector in 3-dimensional space is defined as an equivalence class under this equivalence relation; a vector in 3-dimensional space is therefore the set of all directed line segments in 3-dimensional space that have the same magnitude and direction.

Any vector in 3-dimensional space can be represented as an ordered triple of real numbers as follows:

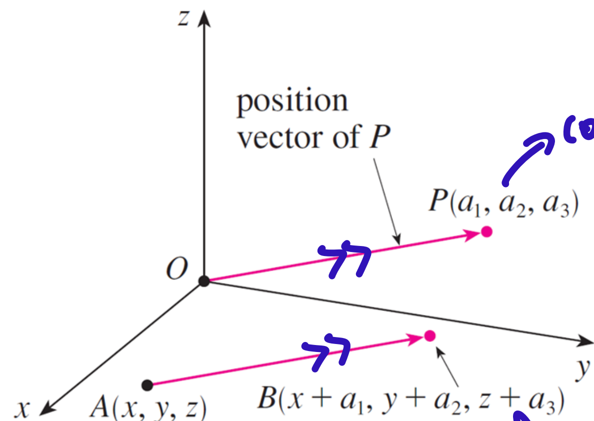
- Let $\mathbf{u}$ be a vector in 3-dimensional space.
- Euclidean Geometry says that we can find a unique point P in 3-dimensional space such that $\mathbf{u}$ contains the directed line segment from O to P.
- Then represent $\mathbf{u}$ as the Cartesian coordinate triple of P.

Displacement Vectors and Position Vectors

→ represents the vector from O to P

$\overrightarrow{OP} = (a_1, a_2, a_3)$

coordinates of P



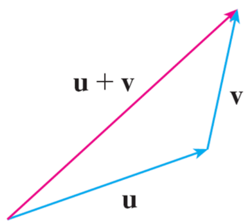
$\overrightarrow{AB} = (a_1, a_2, a_3)$

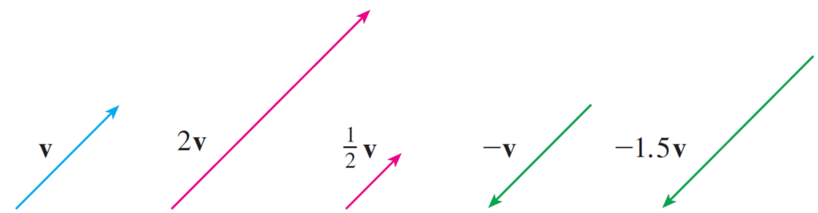
displacement vector from A to B

Given points A and B in 3-dimensional space, we define the displacement vector from A to B to be the vector in 3-dimensional space represented by the directed line segment from A to B, and we denote it by $\overrightarrow{AB}$.

Given a point P in 3-dimensional space, we define the position vector of P to be the displacement vector $\overrightarrow{OP}$.

Vector Addition and Scalar Multiplication



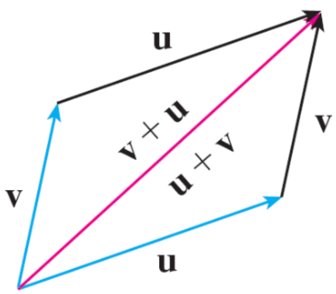


Vector addition is defined by the Triangle Law: → Tells us how to add vectors

- Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in 3-dimensional space.
- Let A and B be points in 3-dimensional space such that $\mathbf{u} = \overrightarrow{AB}$.
- Let C be a point in 3-dimensional space such that $\mathbf{v} = \overrightarrow{BC}$.
- Then we define $\mathbf{u} + \mathbf{v}$ to be $\overrightarrow{AC}$.

Scalar multiplication of vectors is defined by scaling, which is a well-known operation in Euclidean Geometry.

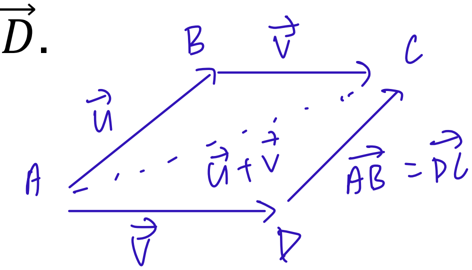
The Parallelogram Law of Vector Addition



u+v = v+u

The Parallelogram Law implies that vector addition is commutative:

- Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in 3-dimensional space.
- Let A, B, and C be points in 3-dimensional space such that $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{BC}$.
- Let D be a point in 3-dimensional space such that $\mathbf{v} = \overrightarrow{AD}$.
- As $\overrightarrow{AD} = \overrightarrow{BC}$, the Parallelogram Law yields $\overrightarrow{AB} = \overrightarrow{DC}$.
- Therefore, $\mathbf{u} + \mathbf{v} = \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{DC} = \mathbf{v} + \mathbf{u}$.



$$\left.\begin{aligned}\vec{u} + \vec{v} &= \overrightarrow{AB} + \overrightarrow{BC} \\ &= \overrightarrow{AC} \\ &= \overrightarrow{AD} + \overrightarrow{DC} \\ &= \vec{v} + \vec{u}\end{aligned}\right\}$$ Parallelogram Law proves that vector addition is commutative

The Algebraic Properties of Vector Operations

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in 3-dimensional space. Let s and t be real scalars.

1. Commutativity: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
2. Associativity: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
3. Existence of the additive identity: There is a unique vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
4. Existence of an additive inverse: There is a unique vector $-\mathbf{u}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
5. Distributivity of vector addition w.r.t. scalar multiplication: $s \cdot (\mathbf{u} + \mathbf{v}) = (s \cdot \mathbf{u}) + (s \cdot \mathbf{v})$.
6. Distributivity of scalar addition w.r.t. scalar multiplication: $(s + t) \cdot \mathbf{u} = (s \cdot \mathbf{u}) + (t \cdot \mathbf{u})$.
7. Compatibility of scalar multiplication with field multiplication: $s \cdot (t \cdot \mathbf{u}) = st \cdot \mathbf{u}$.
8. The identity of scalar multiplication: $1 \cdot \mathbf{u} = \mathbf{u}$.

The Length of a Vector and the Dot Product

Let $\mathbf{u} = (a, b)$ be a vector in 2-dimensional space.

We define the length of $\mathbf{u}$ to be $\|\mathbf{u}\|_2 \coloneqq \sqrt{a^2 + b^2}$. (The 2-D Pythagorean Formula)

Let $\mathbf{u} = (a, b, c)$ be a vector in 3-dimensional space.

We define the length of $\mathbf{u}$ to be $\|\mathbf{u}\|_2 \coloneqq \sqrt{a^2 + b^2 + c^2}$. (The 3-D Pythagorean Formula)

Let $\mathbf{u} = (a, b)$ and $\mathbf{v} = (c, d)$ be vectors in 2-dimensional space.

We define the dot product of $\mathbf{u}$ with $\mathbf{v}$ to be $\mathbf{u} \bullet \mathbf{v} \coloneqq ac + bd$.

Dot product takes 2 vectors as input and give real no. as output

Let $\mathbf{u} = (a, b, c)$ and $\mathbf{v} = (d, e, f)$ be vectors in 3-dimensional space.

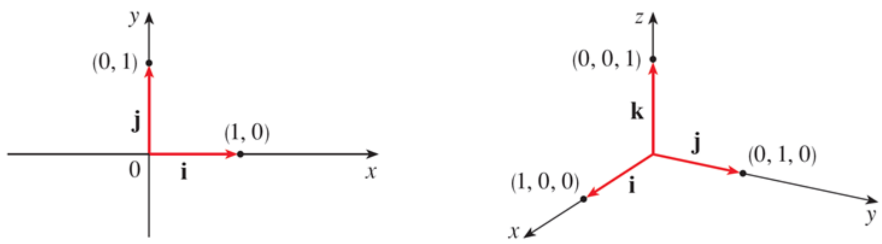
We define the dot product of $\mathbf{u}$ with $\mathbf{v}$ to be $\mathbf{u} \bullet \mathbf{v} \coloneqq ad + be + cf$.

Theorem: Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in 2- or 3-dimensional space.

Then $\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos(\theta)$, where θ denotes the interior angle between $\mathbf{u}$ and $\mathbf{v}$.

$0 \leq \theta \leq \pi$

The Standard Basis Vectors



(a) (b)

Each vector in 2-dimensional space is a linear combination of the standard basis vectors $\mathbf{i}$ and $\mathbf{j}$:

$$(a, b) = (a, 0) + (0, b) = a \cdot (1,0) + b \cdot (0,1) = a \cdot \mathbf{i} + b \cdot \mathbf{j}.$$

Each vector in 3-dimensional space is a linear combination of the standard basis vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$:

$$\begin{aligned}(a, b, c) &= (a, 0,0) + (0, b, 0) + (0,0, c)\\ &= a \cdot (1,0,0) + b \cdot (0,1,0) + c \cdot (0,0,1)\\ &= a \cdot \mathbf{i} + b \cdot \mathbf{j} + c \cdot \mathbf{k}.\end{aligned}$$